

Strength Training for Ultrarunning

In the first edition of this book, I made an egregious error regarding strength training and cross-training. I had a whole section like the one that follows laid out, only to leave it on the cutting room floor and replace it with a small sidebar. The reader reviews stemming from that omission were brutal. While the conclusions made in that sidebar are almost identical to the section that follows, I regret not taking the time to expand fully on why strength and cross-training may be good, how to implement them, and when they might be counterproductive for you.

Some might say that I think strength training is a waste of time, and based on the previous rendition of this book, that's a fair critique. Honestly, we paint the term "strength training" with an overly broad brush. It can mean anything from bar-bending Olympic-style lifts to benign glute activation exercises. Thus, there should be some reasonable calculus to deciding the ultimate goal of strength training for a specific athlete, what exercises to incorporate into that athlete's regimen, and how they are going to incorporate the exercises into their training and lifestyle.

For the purposes of this book, I am going to divide strength training into two distinct categories: "strength training" and "corrective exercise training." In the most basic sense, "strength training" is training for strength. You are literally training in a manner that makes your muscles and tendons stronger so that they will produce more force. This is primarily accomplished by resistance training with progressively heavier loads in order to improve muscle strength, power, hypertrophy, and/or endurance. "Corrective exercise training," on the other hand, is focused on correcting some sort of imbalance or movement pattern, primarily through the use of lighter weights, resistance bands, body weight, and unstable surfaces. While there is some overlap between the two, they are distinct in the fact that strength training will leave you fatigued and corrective exercises, while taxing to specific muscle groups, usually will not. This has programming considerations that will be discussed in short order.

Strength Training as an Ergogenic Aid

Strength training has long been used by long-distance and marathon runners as a way of improving performance and preventing injury (which has the tangential benefit of improving performance through increased training availability). It is a universally accepted practice at the high school, collegiate, and post-collegiate levels, and for good reason. Many studies have looked at the performance benefits of concurrent strength and endurance training in endurance athletes. A quick PubMed search of "strength training and endurance" will yield hundreds of results, all of which add their own twist on prescription, timing, duration, and outcome measurements. Nearly universally, these studies show improvements in running economy (the steady state rate of oxygen you consume and utilize at submaximal pace), increases in VO2 max, improvements in realworld performance, and reductions in injury rates (Balsalobre-Fernández et al. 2016).

However, even with a universally accepted practice, a closer look is always warranted. How big are these improvements? A few percent, maybe. Are they applicable to real-life situations? Probably not. Do they come at a cost of time and effort? Usually. How important are the improvements for ultramarathon performance? That's the question we are after.

STRENGTH TRAINING AND RUNNING ECONOMY

One of the primary benefits lauded by strength-training advocates is the improvement in running economy. Traditional strength training should improve your running economy by improving spring stiffness (the ligaments and tendons) in your legs and via better neuromuscular recruitment. Essentially your legs are springs, and you propel your body with those springs. Some of this movement is the energy needed for muscular contraction, but a lot of that energy is also “free.” Your muscles and tendons stretch during your gait cycle, like a rubber band being stretched, and that elastic energy is harnessed before being released again. The spring stiffness you acquire from jumping and moving heavy weights (creating a thicker rubber band) improves how effectively you can push against the ground as you propel your body forward on the trail. Awesome, right? Well, potentially, but strength training can mean a lot of work for a 2 to 4 percent improvement in running economy (Balsalobre-Fernández et al. 2016), and there is still uncertainty about whether it will matter over the course of a 100-mile effort.

STRENGTH TRAINING AND INJURY REDUCTION

A compelling reduction in injury rates is another big potential benefit of strength training for endurance, according to the scientific literature. One meta-analysis (Lauersen 2018) suggests that athletes who strength train experience a ~66 percent reduction in injury risk. This is important when we factor in training consistency over the long haul, which is a predictor in ultramarathon performance and finishing rates. The calculus is simple: if you miss fewer days of training every year from injuries, you can train more, and more training generally (not always) improves performance. However, there are a lot of other things that lower injury risk, including rest days (Ristolainen et al. 2014), sleeping eight or more hours a night (O’Donnell 2018), and avoiding RED-S. The point is that more is not always better, and when you add something to your workload, that often means having to take something else away.

So, where does this fit into your ultramarathon training? Unfortunately, there is not a simple binary “Yes, do it!” or “No, don’t do it!” answer. Just like any piece of your training puzzle, individuality is key. As mentioned above, this kind of strength training is hard, meaning there is a demand on the body and subsequently a higher overall workload or training stress. In nearly all cases, adding this kind of training means eliminating another quality session during your week, like that second interval session. There are times of the year when that is of much less consequence and might even have a lot of benefit. What that means is it’s important to look at your LRP, decide on the most important aspects of your training goals for each block, and ask yourself, “What do I prioritize, and what do I sacrifice?”

ARE YOU A GOOD CANDIDATE FOR STRENGTH TRAINING?

Not all ultramarathon runners are good candidates for strength training. While there are gains to be made in the weight room, there are also gains to be made out on the trails. If you are looking at improving your race time or finishing chances by a few percent, you can choose to do so by adding more miles, incorporating more recovery, or hitting the gym, but not necessarily all three, because they will be mutually exclusive at times. For example, if you decide to add two hours of strength training per week, that’s two hours you could have spent simply resting. If you have enough rest and recovery time built into your schedule, that might not be a big deal. But, if you are burning the training candle at both ends, you might be better off just taking that time to lay on the couch. Time and energy are limited commodities, even for elite athletes, so you must choose wisely.

When I work with athletes, I go through a flowchart similar to Figure 11.1 to determine if they are good candidates for strength training. After completing this exercise, some are good candidates, but many are not.

PRESCRIPTION FOR STRENGTH TRAINING

You've determined that you're a candidate for strength training—great! The key now is to develop a smart and efficient program that supports your running volume and intensity while allowing for appropriate recovery. Although the reps and sets (volume) will change depending on where you are in your training phase, there will always be five basic movement patterns you should perform:

1. PUSH: includes exercises like push-ups, bench press, and other types of upper body push presses
2. PULL: includes exercises like bent-over rows and pull-ups
3. HINGE (OR HIP HINGE): includes exercises like a Romanian deadlift
4. SQUAT: includes exercises like a squat (go figure), lunge, or reverse lunge
5. CARRY (OR OTHER TYPES OF DYNAMIC CORE WORK): includes a farmer's carry, woodchoppers, and medicine ball slams

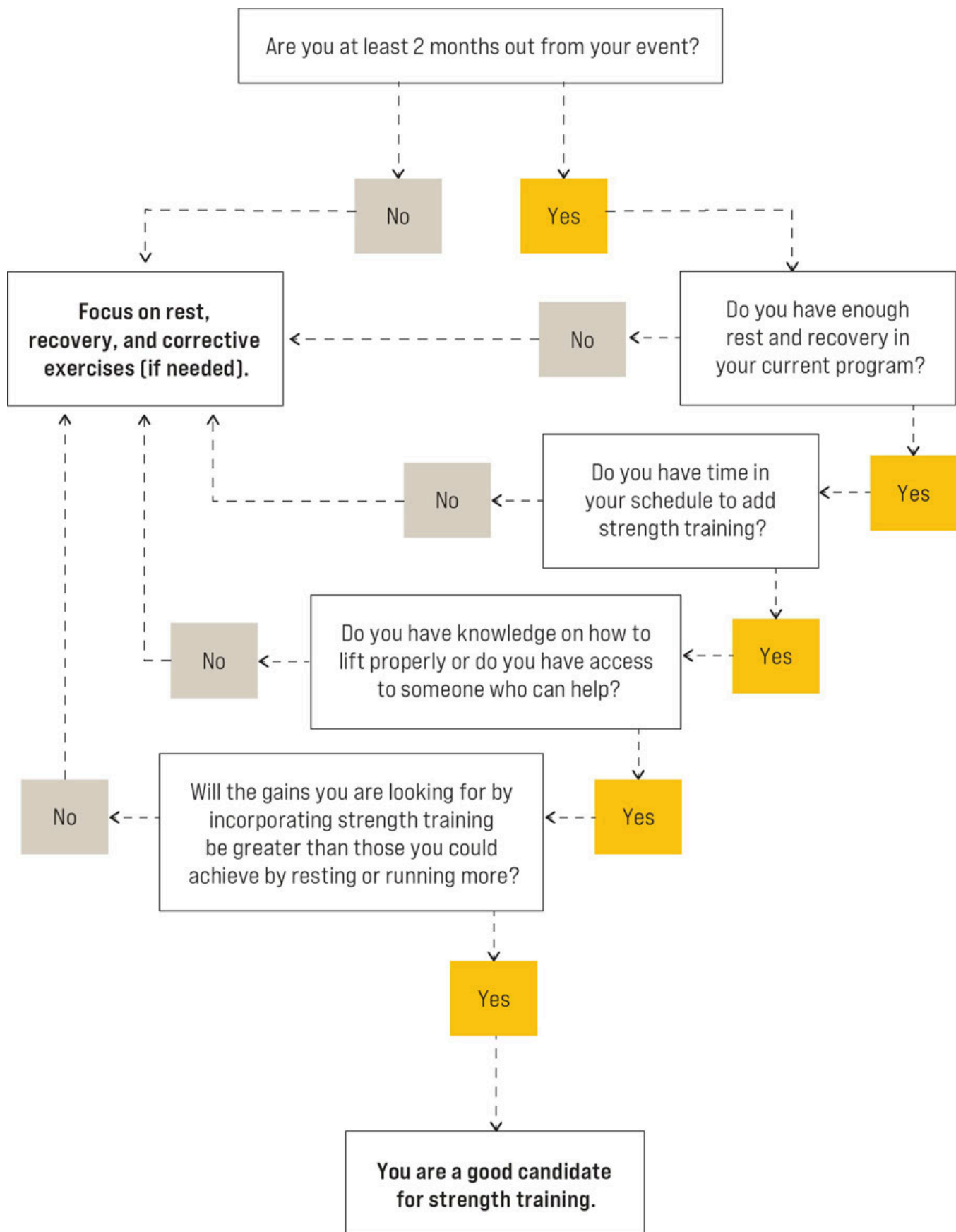


FIGURE 11.1 Are you a good candidate for strength training?

These exercises can be organized into a two- or three-days-per-week strength program. An example follows, using Day A as the first day and Day B as the second. If you choose to have a three-day-per-week strength program, you can repeat your preference of either Day A or B.

Day A

- PUSH-UP
- DUMBBELL ROW
- ROMANIAN DEADLIFT
- SQUAT
- FARMER'S CARRY

Day B

- OVERHEAD PRESS
- PULL-UP (WITH ASSISTANCE IF NEEDED)
- HIP THRUST
- REVERSE LUNGE
- MEDICINE BALL SLAM

Now, how do we structure these training days into the overall architecture of your weekly schedule? Remember that strength training should complement your run training, not detract from it.

Looking at the calendar in Figure 11.2, you will notice a few things. First you will see that your strength days are scheduled as far away as possible from your next long run and/or hard runs. This will allow you to go into those workouts as fresh as possible so you can have the best workout. In addition, pairing your strength-training days on the evenings of where more intense work is programmed allows you to truly take your easy days easy. On the

days when you lift and run on the same day, it is important to prioritize your run training, taking care to program your runs prior to your strength work.

For your particular schedule, you might be forced to do your long runs in the evenings or on a different day, but in any case, the overall strategy of placing your strength- training days such that you have as much recovery as possible should reign supreme. Things get particularly tricky if you add a third strength-training day to the week, and normally I don’t advise that for athletes focusing on ultramarathon performance.



FIGURE 11.2 Example training week with two scheduled strength-training workouts.

ORGANIZING STRENGTH TRAINING THROUGHOUT THE SEASON

When determining how often and how much strength training to do within your running season or year, the best advice is to look at where you are in your run programming and then see where your strength training lines up and fits in. Remember, your priority here is to achieve appropriate run volume balanced with recovery, and strength training should complement this balance of training and recovery, never detract from it.

With that in mind, a good guideline is to look at the entire season of training before your event as a big pie, and divide the pie into three segments: an early season, a mid season, and a late season. For the purpose of this example we are going to assume that you begin your training in January for a race in September, so you have nine months to train. The early season is the very beginning of your training, and this is usually months 1 to 3 of the pie. Your mid season would be months 4 to 7, with the late season (the last two months before your event!) being months 8 and 9.

PART OF TRAINING	SETS	REPS	REST	FREQUENCY
Early	1-5	1-5	90+ sec	3x week
Mid	3	8-12	45-60 sec	2-3x week
Late	2-3	10-15 or time-based	As needed	1-2x week

TABLE 11.1 Overview of strength-training programming.

The Early Season

The early season is before either run volume or intensity has increased dramatically. This is when you build the foundations for the coming phases. In some cases, you start the early season by coming off a bit of a break or a period of decreased run volume. In this season of running, your strength training should be made up of one to five sets of one to five reps of big compound movements utilizing heavier weight, with longer recoveries between sets. The focus here is heavier weights and lower reps and sets. Because run volume is nowhere near its peak, you have more time to balance the stress and load of strength training with your running while still having adequate time to recover.

The Mid Season

The mid season is when you will start to see an increase in intensity or volume of running, depending on your training block. In any case, your run training will be harder and consume more of your physiological bandwidth. Because of this, strength training will begin to emphasize moderate sets and reps, with a goal of three sets of eight to twelve reps, lifting weight that is a challenge but leaving one or two reps in the tank— so not to failure.

The Late Season

At this point you are zeroing in on your event, and all of your body's resources should be dedicated to the specific demands of that event. Rest and recovery days are at a premium, and you want to reduce or eliminate adding any unnecessary stress to the body. Therefore, this is the phase to concentrate more on corrective and restorative exercises (see next section). This type of work is minimally taxing on both your body and your time. It is a time to shore up any areas of weakness, work on mobility, and allow for more recovery in the place of longer, more intense strength-training sessions. Higher reps are more appropriate with the exercise selection and can also be time-based (i.e., thirty seconds per exercise or stretch). The programming of these sessions should be one to two days a week of work that complement your run training.

In all of these prescriptions, the set/rep range should be ~70 to 80 percent of what you feel is your maximum. For example, on a set of five reps, you should feel like you could do one or two more repetitions before failure.

Corrective Exercise Training

Corrective strength training is perhaps the most common form of strength work we see in the endurance community. Akin to physical therapy, the goal of a corrective exercise program is to target specific movements that address minor weaknesses and imbalances particular to your anatomy, movement patterns, and strength. This style of training has a low cost of effort and is less time intensive compared to traditional strength training. Exercises that fall into this category generally include body-weight exercises, clamshells, banded walking drills, and exercises done on balance balls or other unstable surfaces. These exercises potentially protect athletes from injury by correcting imbalances, but it's unclear if this style of training could influence injury risk in the same way as traditional strength training. The benefit of this style of training is that it has little impact on the quality of focused interval workouts and long runs. This means corrective exercise training is a low-risk addition to your training cycle that "could" have the added benefit of minor injury prevention.

ARE YOU A GOOD CANDIDATE FOR CORRECTIVE EXERCISES?

Being that most corrective exercise programs have little to no impact on higher-quality run sessions, most runners would assume that everyone is a candidate. After all, if there is no harm, you might as well do it, right? Still, there is a time cost involved with incorporating even the most basic corrective exercise program. If you have the ability to overcome this time cost, a simple corrective exercise program could be for you. More importantly, if you are a runner who is constantly injured, investing time in a corrective exercise program, along with managing your overall running workload, is essential. In these cases, I suggest seeking out and working with a qualified physical therapist.

WHAT ABOUT LEG CIRCUITS AND BODY-WEIGHT EXERCISES?

STRENGTH TRAINING HAS ALWAYS BEEN a tricky element to program for ultramarathon athletes. High running volume and copious amounts of climbing and descending leave trail runners with tired legs on a daily basis. The last thing many of them want to do is add additional workload to their legs, particularly if it's hard and heavy (as strength training should be). This has led many coaches and athletes to come up with an "in-between" solution using leg circuits, body-weight squats, and other high-volume/low-weight programs. Some of these are even timed, have clever names, and are encouraged to be performed daily.

High-repetition and low-weight strength-training exercises work only very temporarily and primarily in a corrective setting. Just like run training, strength training should be purposeful. It should be hard and require recovery in order to reap adaptation. The issue with frequent high-repetition leg-circuit exercises is that after the initial adaptation phase (usually just a few short weeks), there is no further progress to be had, as there is no source of progressive overload. It is literally the worst of both worlds, as the resistance is not high enough to elicit a strength adaptation, and the cardiopulmonary stress is not specific enough to translate into running performance. Additionally, in the cases where the circuits are timed, the emphasis is put on speed or volume because the number of repetitions performed serves as a score that needs to be improved. This sacrifices technique and increases the likelihood of injury. So, even if you are crunched for time, you are better off redirecting your efforts into a few heavy squats as opposed to several dozen body-weight squats, lunges, and matrixes.

PRESCRIPTION FOR CORRECTIVE EXERCISE TRAINING

When it comes to corrective exercise training there are many areas runners in particular can benefit from, including:

- Posterior chain exercises for the glutes, hips, and hamstrings: These can include ball bridges, traditional bridges, and banded glute work.
- Exercises to strengthen weak feet: These can include barefoot balance work, foot stretches and activation, and top-of-the-foot and calf stretches.

- Exercises for hips and pelvic stability: These can include monster band walks, lateral step-ups, and the pelvic list.

Most of these exercises can be done on alternating days throughout the week, up to three days per week. Work up to completing two to three sets of up to twelve reps or fifteen to thirty seconds. If you are just starting out, pick one from each category and rotate the exercises each session.

But Won't I Gain Weight That Will Slow Me Down?

The most common concern I hear from endurance athletes is often the worry that they'll bulk up from strength training. The reality is few of the endurance athletes reading this book are going to look like a professional bodybuilder from a couple of days per week of strength training. In most cases, there is nearly no muscle hypertrophy when performing concurrent strength and endurance training. In fact, in long-term studies (twenty-four to forty weeks) looking at concurrent strength and endurance training, there were no signs of muscular growth despite improvements in strength measurements (Blagrow, Howatson, and Hays 2017). This happens because of what occurs at the molecular level when practicing concurrent training. On one hand, strength training induces an upregulation in protein synthesis and releases a specific protein, called mTOR, that in a normal strength-training setting would help you build your beach muscles. However, when concurrent endurance training is taking place, you release an enzyme called adenosine monophosphate-activated kinase that blunts the effects of mTOR. What this means is that as an endurance athlete it is possible to increase muscular strength, your ability to generate power, and your muscle-tendon stiffness without the potentially negative side effects of mass gain.

Aging Athlete Considerations

Aging athletes are in a battle against the natural age-related loss of muscle mass that starts somewhere north of 40 and accelerates rapidly as you cross the 60-year threshold. From a performance standpoint we also know that this decrease in muscle mass is one of the main culprits when it comes to declining performance, as age-related decreases in VO2 max are directly related to the loss of muscle mass. So, for aging athletes, strength training isn't just corrective or ergogenic, it's essential for general health and well-being. Trying to build and maintain muscle mass is of utmost importance and should be taken into account when you look at your training availability each week. For younger athletes, strength training might take a back seat for much of the year, but as we age we are all encouraged to shift some priority away from running time and toward moving heavy objects a few times a week.

A Place for Cross-Training

If you only have time to run, you should be spending that time running, not cross-training. Yoga, CrossFit, Zumba, and cycling might make you a better, more well-rounded athlete, but will it make you a better ultramarathoner? Probably not. Running is what will make you better at running.

However, there may be a time and place for cross-training sometime in the course of a year or season. Utilizing other training modalities can make sense when you can't run due to an active injury, when you're coming back from injury, or when your goal races are far off. Additionally, if you are a workload-capped runner (you have more available time to train, but adding more running workload leads to injury), using a bike, skis, or a pool to add training can be effective. Furthermore, if you simply like getting on your bike once or twice a week, go ahead

and give it a whirl. Just don't think that it will automatically make you a better runner. The same goes for yoga, swimming, rowing, and the like: if you enjoy doing these activities, then great! But in the vast majority of cases, translating those activities to being a better, faster trail runner is a stretch. From a physiological standpoint, fitness and specificity of training are the most important things come race day, so keep the big picture in mind if you choose to supplement your training with nonrunning activities.

I would like to thank Sarah Scozzaro for her contributions with this chapter.

KEY POINTS FOR STRENGTH TRAINING FOR ULTRARUNNERS

Ultrarunners can use strength training to acutely enhance their performance through running economy and/or injury reduction.

Not all ultrarunners are candidates for strength training.

Strength training comes at a compromise of time and energy, which must be balanced depending on your goals.

Corrective exercise training is a low-cost, low-effort training adjunctive that can be used to mitigate injury risk.

If you like other forms of cross-training, that's great. Just don't expect them to improve your ultrarunning performance in any meaningful way.



